Question 1: An RGB color image has a spatial resolution of 640 × 480 pixels.

Each pixel is represented using 8 bits per color channel (R, G, and B).

1. Convert the given RGB pixel (R = 120, G = 200, B = 80) into its equivalent grayscale value using the standard luminance equation. (3 marks)

2. Convert the same RGB pixel (R = 120, G = 200, B = 80) into the HSV color model.

Clearly show the calculation of Hue, Saturation, and Value. (4 marks)

3. Calculate the total size of the RGB image in megabytes (MB) before compression. (2 marks)

4. After applying an image compression technique, the image size is reduced to 450 KB.

Calculate the compression ratio. (2 marks)

5. If the image is transmitted at 25 frames per second,

determine the required bit rate for transmission of the compressed image. (1 mark)

Question 2:A grayscale digital image has a spatial resolution of 1024 × 768 pixels and is represented using 6 bits per pixel.

1. Calculate the total number of pixels in the image. (2 marks)

2. Determine the gray level resolution of the image and specify the intensity range. (2 marks)

3. Calculate the total memory required to store the image in kilobytes (KB) and megabytes (MB). (3 marks)

4. The image is obtained by sampling a continuous-time image signal having a maximum frequency of 4 kHz.

• Check whether a sampling frequency of 7 kHz will cause aliasing. Justify your answer using the sampling theorem. (3 marks)

5. Explain 4-connected, 8-connected, and m-connected pixel connectivity with neat diagrams. (2 marks)

Question 3:

Consider the following two discrete-time signals:

$x(n) = \{1,\ 2,\ 1\}$

$h(n) = \{2,\ 1,\ -1\}$

1. **Compute the linear convolution** of $x(n)$ and $h(n)$. Show all steps. *(3 marks)*
2. **Find the 3-point circular convolution** of $x(n)$ and $h(n)$. *(2 marks)*
3. **Calculate the auto-correlation** of the signal $x(n)$. *(2 marks)*
4. A **3×3 grayscale image segment** is given below:

$$\begin{bmatrix} 100 & 120 & 140 \\ 110 & 130 & 150 \\ 120 & 140 & 160 \end{bmatrix}$$

Apply a **3×3 mean filter** and determine the **output value of the center pixel**. *(2 marks)*

5. The histogram of a grayscale image has **64 gray levels** and **4096 total pixels** uniformly distributed.
   - **Perform histogram equalization** and find the **new gray level corresponding to original gray level 20**. *(2 marks)*
6. **State one difference between linear convolution and circular convolution.** *(1 mark)*

Question 4:Design a digital low-pass FIR filter **using the** window method **to meet the following specifications:**

- Passband edge frequency, f_p = 1 kHz
- Stopband edge frequency, f_s = 1.5 kHz
- Sampling frequency, f_{sampling} = 8 kHz
- Minimum stopband attenuation = **40 dB**

1. **Determine the normalized digital passband and stopband frequencies.** *(2 marks)*
2. **Select a suitable window function** to satisfy the given attenuation requirement and justify your choice. *(2 marks)*
3. **Calculate the transition width and estimate the order of the FIR filter.** *(3 marks)*
4. **Write the expression for the ideal impulse response of the low-pass filter.** *(2 marks)*
5. **Explain how the selected window modifies the ideal impulse response to obtain the practical FIR filter.** *(2 marks)*
6. **State one advantage of FIR filters over IIR filters.** *(1 mark)*

Question 5:

A binary image A and a structuring element B are given below:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

1. **Perform the dilation operation** $A \oplus B$ **and obtain the output image.** ***(3 marks)***
2. **Perform the erosion operation** $A \ominus B$ **and obtain the output image.** ***(3 marks)***
3. **Find the opening of the image** $A \circ B$. ***(2 marks)***
4. **Find the closing of the image** $A \bullet B$. ***(2 marks)***
5. **Explain boundary extraction and write the expression used for boundary extraction in morphological processing.** ***(1 mark)***
6. **State one application of morphological image processing in segmentation.** ***(1 mark)***

Question 6:

Consider the discrete-time signal

x(n) = {1,0,2,0\}, n = 0,1,2,3

1. **Compute the 4-point Discrete Fourier Transform (DFT)** of the signal x(n).

Show all steps. *(4 marks)*

2. **Find the 4-point Inverse Discrete Fourier Transform (IDFT)** of the result obtained in part (1) and verify that the original signal is recovered. *(2 marks)*
3. **Perform the 4-point Fast Fourier Transform (FFT)** of x(n) using the **Decimation-in-Time (DIT)** algorithm.

   Draw the butterfly diagram and compute the final FFT output. *(3 marks)*

4. **Verify the linearity property of the DFT** using the signals

   x_1(n) = {1,0,1,0} and

   x_2(n) = {0,1,0,1}. *(2 marks)*

5. **State one advantage of FFT over direct DFT computation.** *(1 mark)*